

CMP-23 - State and Persistence Platform

Provide durable authoritative storage primitives, concurrency controls, snapshots, projections, replicas, and migration facilities while preserving domain ownership.

Authority and state ownership

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 24. Authoritative state: storage engines, commit logs, snapshots, replication metadata, projection and migration state. Explicitly non-owned: semantic ownership of domain records, policy decisions, evidence meaning, global transaction authority.

Critical invariant

Each authoritative state domain has exactly one mutation owner; replicas, caches, and projections remain explicitly derived.

Logical interfaces

CMP-23-IF-01 / Command / Commit State Change

Command `Commit State Change` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-02 / Command / Create Snapshot

Command `Create Snapshot` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-03 / Command / Migrate Store

Command `Migrate Store` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-04 / Command / Rebuild Projection

Command `Rebuild Projection` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-05 / Query / Read State

Query `Read State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-06 / Query / Get Version

Query `Get Version` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-07 / Query / Verify Replica

Query `Verify Replica` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-08 / Query / Get Migration State

Query `Get Migration State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-09 / Event / State Committed

Event `State Committed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-10 / Event / Snapshot Created

Event `Snapshot Created` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-11 / Event / Migration Completed

Event `Migration Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-23-IF-12 / Event / Replica Diverged

Event `Replica Diverged` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Provisioning > Ready > Degraded > Recovering > Ready or Retired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: split brain, lost update, snapshot corruption, migration partial failure, projection treated as authority. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-23-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-23-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-23-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-23-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-23-TST-005 - Critical invariant. Each authoritative state domain has exactly one mutation owner; replicas, caches, and projections remain explicitly derived. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-23-TST-006 - Failure containment. Inject split brain, lost update, snapshot corruption, migration partial failure, projection treated as authority. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-23-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-23-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-23-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-23-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.